

VASCO XU

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EDUCATION

University of Chicago

2021 - Present

Ph.D (& M.S) in Computer Science

Advisors: Dr. Hank Hoffmann & Dr. Karan Ahuja (Northwestern University)

University of Pittsburgh

2016 - 2020

B.Phil (& B.S) in Computer Science

Advisors: Dr. Sherif Khattab & Dr. Daniel Mossé

PROFESSIONAL EXPERIENCE

Google

Jun. 2026 - Now

PhD Software Engineer Intern

San Jose, CA

- Developing ultra-lightweight speech enhancement models for assistive listening on AR glasses.
- Mentors: Khushman Patel and Anish Prabhu.

Google

Apr. 2024 - Sept. 2024

Student Researcher, AR Perception

Seattle, WA

- Developed XR interaction systems for low-vision accessibility, surface-bounded interactions, and text entry.
- Mentors: Mar Gonzalez-Franco and Harish Kulkarni.

Immunetrics

Aug. 2019 - Dec. 2019

Software Support Engineer

Pittsburgh, PA

- Developed a web application for a biological model building platform.

PUBLICATIONS

- [11] **ArmPoser: Real-Time, Calibration-Free Arm Pose Estimation from Smartwatch IMU**
Bishnu Dev, **Vasco Xu**, Xi-Aan Loh, Chenfeng Gao, Henry Hoffmann, and Karan Ahuja
SUI 2026 – *ACM Symposium on Spatial User Interaction*
- [10] **FlowAvatar: Real-Time Full-Body Avatars from Sparse Egocentric Inputs on Consumer XR Devices**
Chenfeng Gao, Taeyoung Yeon, Sungheon Park, **Vasco Xu**, Anish Prabhu, Mar Gonzalez-Franco, and Karan Ahuja
ISMAR 2026 – *IEEE International Symposium on Mixed and Augmented Reality*
- [9] **MARIO: Motion-Augmented Real-Time Multi-Sensor Inertial Odometry**
Yiquan Li, Taeyoung Yeon, Chenfeng Gao, **Vasco Xu**, Xuanyou Liu, and Karan Ahuja
CVPR 2026 [Findings] – *Conference on Computer Vision and Pattern Recognition*
- [8] **SurfaceXR: Fusing Smartwatch IMUs and Ego-centric Hand Pose for Seamless Surface Interactions**
Vasco Xu, Brian Chen, Eric J Gonzalez, Andrea Colaço, Henry Hoffmann, Mar Gonzalez-Franco, and Karan Ahuja
IEEE VR 2026 – *IEEE Conference on Virtual Reality and 3D User Interfaces (TVCG Special Issue)*
- [7] **Gambler: Adaptive Sampling Under Energy Budgets**
Vasco Xu, Tejas Kannan, and Henry Hoffmann
EWSN 2025 – *ACM International Conference on Embedded Wireless Systems and Networks*

- [6] **WatchHAR: Real-time On-device Human Activity Recognition System for Smartwatches**
Taeyoung Yeon, **Vasco Xu**, Henry Hoffmann, and Karan Ahuja
ICMI 2025 – *ACM International Conference on Multimodal Interaction*
- [5] **Multi-dimensional Engagement for Richer Hyperlocal Citizen Science Data**
Abhishek Viswanathan, **Vasco Xu**, Amy Babay, Rosta Farzan, and Aaron Birdy
C&T 2025 – *International Conference on Communities & Technologies*
- [4] **MobilePoser: Real-Time Full-Body Pose Estimation and 3D Human Translation from IMUs in Mobile Consumer Devices**
Vasco Xu, Chenfeng Gao, Henry Hoffmann, and Karan Ahuja
UIST 2024 – *ACM Symposium on User Interface Software and Technology*
- [3] **MotionTrace: IMU-based Field of View Prediction for Smartphone AR Interactions**
Rahul Islam, **Vasco Xu**, and Karan Ahuja
BSN 2024 – *IEEE-EMBS International Conference on Body Sensor Networks*
- [2] **Xs: Interactive Scissor Mechanisms as Portable and Customizable Shape-Changing Interfaces**
Vasco Xu and Ken Nakagaki
TEI 2023 – *ACM International Conference on Tangible, Embedded and Embodied Interaction*
- [1] **LUSH: Lightweight Framework For User-Level Scheduling in Heterogeneous Multicores**
Vasco Xu, Liam White McShane, and Daniel Mossé
MCSoc 2021 – *IEEE International Symposium on Embedded Multicore/Many-core Systems-on-Chip*

PRESENTATIONS AND EXHIBITIONS

- 7. [**Meta Reality Labs**] “INRTL: Inertial Real-time Tracking and Localization”, 2026
- 6. [**Qualcomm XR**] “INRTL: Inertial Real-time Tracking and Localization”, 2025
- 5. [**Google AR**] “Surface Plane Interactions”, 2025
- 4. [**South by Southwest (SXSW)**] “Xs Sculpture”, 2024
- 3. [**Chicago South Side Science Festival**] “Xs Sculpture”, 2023
- 2. [**Ars Electronica**] “Xs Sculpture”, 2023
- 1. [**Symposium on the Science of Security (HotSoS)**] “MAZE: A Secure Cloud Storage Service Using Moving Target Defense and Secure Shell Protocol (SSH) Tunneling”, 2021

SELECTED HONORS AND AWARDS

Three Minute Thesis (3MT) Finalist , University of Chicago	2026
McCormick Fellowship , University of Chicago	2021
Honorable Mention for Outstanding Undergraduate Researcher , Computing Research Association (CRA)	2021
Bachelor of Philosophy Degree (highest undergraduate distinction) , University of Pittsburgh	2020
Chancellor’s Undergraduate Research Fellowship , University of Pittsburgh	2020
Seed Scholar Grant , University of Pittsburgh	2020
Dietrich School Scholarship , University of Pittsburgh	2017

PAPER REVIEWING

ACM CHI (1 Special Recognition for Outstanding Reviews)	2025, 2026
ACM UIST	2024, 2025, 2026

ACM IMWUT (1 Special Recognition for Outstanding Reviews)	2024, 2025, 2026
ACM ISS	2026
ACM TEI	2025
ACM CHI LBW	2025

STUDENT MENTORING

Bishnu Dev, Masters, MBZUAI	2023-2026
Brian Chen, Undergraduate, Northwestern University	2024

TEACHING EXPERIENCE

• Guest Lecturer, Machine Learning and Sensing Northwestern University	2025, 2026
• Guest Lecturer, AI/ML Bootcamp Google Seattle	2024
• Guest Lecturer, Advanced Topics in Ubiquitous Computing Northwestern University	2024
• Undergraduate Teaching Assistant, Introduction to Programming with Java University of Pittsburgh	2019

SELECTED PRESS COVERAGE

Northwestern Engineering New App Performs Real-Time, Full-Body Motion Capture with a Mobile Device	2024
Hackster.io Tracking Movement on the Move	2024